NEW SCIENTIFIC CONCEPTS PROMISE HEALTH AND LONGEVITY

*MIROSLAV RADMAN**

MEDILS - MEDITERRANEAN INSTITUTE FOR THE STUDY OF LIFE

An international, private, not-for-profit institute in Split (Croatia) is an "investment in surprise" (www.medils.org)

Our research generated two significant concepts:

1. DAMAGED PROTEINS ARE COMMON ROOT CAUSE OF AGING AND AGE-RELATED DISEASES
2. "CELLULAR PARABIOSIS" PREVENTS THE ONSET OF DISEASES BY HEALING SICK CELLS
   The two discoveries can be phrased in an abstract but familiar way: health is biological harmony, disease is disharmony and aging is an increasing noise. With musicians, mathematicians and physicists, I explore ways to measure harmony and identify disharmony. Prevention and healing of diseases amounts to achieve maintenance and reestablishment of natural biochemical harmony seen in healthy young people. The reality of biological rejuvenation was amply documented in so-called heterochronic parabiosis experiments.

Two above concepts translated into simple treatments (e.g., capsules or pills) are expected to result in extension of human lifespan by elimination of common deadly diseases that currently cause about 90% of deaths. This would be a profound change in medicine and public health via new predictive diagnostic of inborn predispositions to diseases, their targeted prevention, and even healing of degenerative and malignant diseases by reversion to health. Such avoidance of common age-related diseases and conditions would prolong healthy and productive life up to 120 years. (With current societal disharmonies and a knowledge-based economy, just how precious could be the excellence and wisdom of biologically young centenarians? It becomes ethical and reasonable to be impatient.)

To propose this redirection of biomedical research, it took 15 years of iconoclast research done "underground" with very modest resources and in necessary isolation from the mighty bio-medical mainstream that - after half a century of the largest global scientific effort of all times - produced expensive "smart drugs" that do not cure any of common age-related diseases. At best and rarely, such drugs ease some of the disease symptoms. "Smart drugs" are targeted to arbitrary molecular targets (called biomarkers) that belong to countless molecular consequences of the biological damage done by the disease. They are not expected to cure diseases. Yet, after half century of failure, billions are yearly poured into the same R&D approach which by common sense is clearly on the wrong track. It is matter of ethics to spell the truth.

Translation to practice of the above two concepts cannot be achieved "underground" since it requires larger resources than the original discoveries. Now, the initial (before clinical trials) financial resources needed count in tens of millions dollars/euros and can come from three sources:
(i) philanthropy, i.e., donations from patrons, (ii) national and supranational governments (if they make decisions about priorities and leadership in global health) and (iii) investments in start-up companies that, being pure business enterprises, often ruin the potential success by the unproductive clash between business and scientific cultures.
Life and health are supreme commodities that should not be left in care of people grounded by nitty-gritty commercial mentality. Thus, this project now depends on visionary Renaissance-style patrons and wise governments (which could make good use of money saved on health budgets).
As to negative economic consequences of the mitigation of age-related diseases and delayed aging feared by unimaginative politicians, I guess that none of them is nostalgic about deaths by infectious diseases in the past. This is matter of life and death! But, even prosaically, business-wise - since nobody desires to become a patient - the entire human population would rather be client consuming molecules that prevent and attenuate diseases. Hence, the largest possible health market.

**M.R. is the founder of MEDILS, Professor Emeritus from University R. Descartes-Paris5 Medical School and member of (among others) French Académie des Sciences, USA National Academy of Sciences and AAAS.*